Typology-Wise Vulnerability Parameters & Cutoff Threshold Report

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1. Cross-Typology Distribution & Threshold Summary

Overview of the statistical parameters (Mean μ, Std Dev σ) fitted via Gaussian distribution and the exact 5th percentile tail threshold cutoff used for noise elimination across all building typologies:

Typology	Structural Description	Total	Mean (μ)	Std Dev (σ)	5% Cutoff	Prominent	Noise
BCBM	Burnt Clay Brick Masonry (BCBM) Typology	5	33.30%	0.00%	**33.30%**	5	0
RC	Reinforced Concrete (RC) Frame Typology	27	9.43%	10.99%	**1.90%**	22	5
RRSM	Random Rubble Stone Masonry (RRSM) Typology	9	6.67%	5.33%	**4.00%**	9	0

2. Typology-Wise Detailed Parameter Lists & Threshold Analysis

Typology: Burnt Clay Brick Masonry (BCBM) Typology (BCBM)

Parameters Evaluated: 5 | **Mean (μ):** 33.30% | **Std Dev (σ):** 0.00% | **5% Threshold Cutoff:** 33.30% | **Core Prominent Retained:** 5

#	Parameter ID	Source	Vulnerability Description / Damage Mechanism	Occurrence	5% Cutoff	Status
1	**BCBM_RVS_4.9**	RVS Primer	Building has cracks in walls.	**33.3%**	33.3%	**Prominent**
2	**BCBM_POST_5.1**	RVS Primer	Building has sustained collapse of cantilevers, balconies, chimneys and parapets.	**33.3%**	33.3%	**Prominent**
3	**BCBM_POST_3.9**	RVS Primer	Building has diagonal shear cracking in wall piers and/or spandrels.	**33.3%**	33.3%	**Prominent**
4	**BCBM_POST_3.3**	RVS Primer	Building has plinth masonry severely damaged.	**33.3%**	33.3%	**Prominent**
5	**ADD_BCBM_01**	Additional	Cyclic shear cracks in industrial brick masonry chimney	**33.3%**	33.3%	**Prominent**

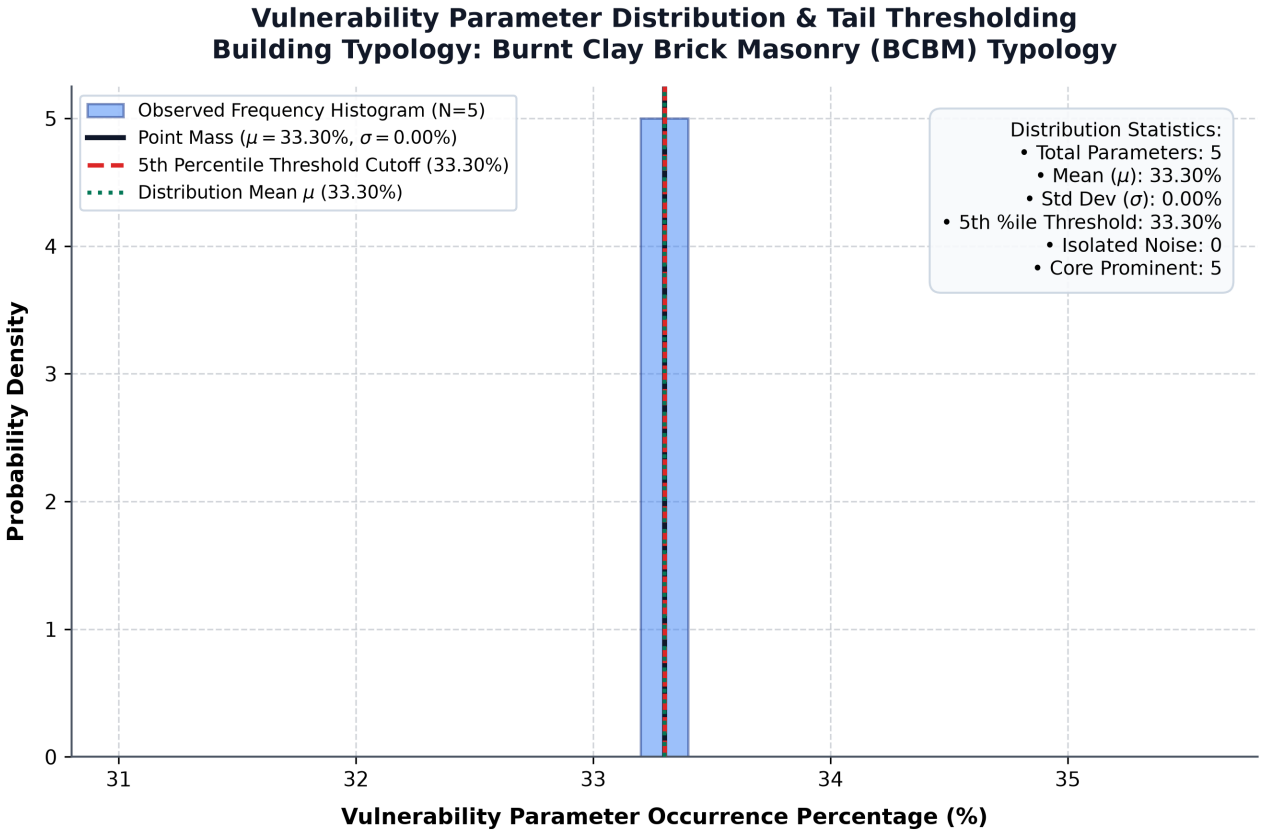


Figure: Normal distribution bell curve & 5% cutoff line for BCBM.

Typology: Reinforced Concrete (RC) Frame Typology (RC)

Parameters Evaluated: 27 | **Mean (μ):** 9.43% | **Std Dev (σ):** 10.99% | **5% Threshold Cutoff:** 1.90% | **Core Prominent Retained:** 22

#	Parameter ID	Source	Vulnerability Description / Damage Mechanism	Occurrence	5% Cutoff	Status
1	**RC_RVS_3.4**	RVS Primer	Building has an Open Storey (at ground /other level) with slender columns.	**45.3%**	1.9%	**Prominent**
2	**RC_POST_3.3**	RVS Primer	Building has an Open storey (at ground /other level) with columns having shear cracks.	**45.3%**	1.9%	**Prominent**
3	**ADD_RC_03**	Additional	Rooftop swimming pool causing extreme mass irregularity and asymmetric torsional collapse	**17.0%**	1.9%	**Prominent**
4	**ADD_GEN_01**	Additional	Common shared wall between adjacent buildings with differential response	**17.0%**	1.9%	**Prominent**
5	**ADD_RC_05**	Additional	Collapse of intermediate storey in commercial RC frame	**15.1%**	1.9%	**Prominent**
6	**RC_RVS_8.2**	RVS Primer	Building has RC walls unsymmetrical in both directions.	**9.4%**	1.9%	**Prominent**
7	**RC_POST_3.9**	RVS Primer	Building has extensive X cracking or out-of-plane collapse of infills.	**9.4%**	1.9%	**Prominent**
8	**ADD_RC_02**	Additional	Lack of core confinement / 90-degree ties with excessive longitudinal spacing	**9.4%**	1.9%	**Prominent**
9	**RC_RVS_9.1**	RVS Primer	Building has longitudinal cracks and/or spalling in Beams and Columns.	**7.5%**	1.9%	**Prominent**
10	**RC_RVS_8.4**	RVS Primer	Building has unanchored water tanks on roof.	**7.5%**	1.9%	**Prominent**
11	**RC_RVS_3.6**	RVS Primer	Building has irregular plan geometry and or plan aspect ratio > 4.0.	**7.5%**	1.9%	**Prominent**
12	**RC_POST_5.2**	RVS Primer	Building has water tanks on roof displaced from their supports.	**7.5%**	1.9%	**Prominent**
13	**ADD_RC_08**	Additional	Improper post-earthquake column jacketing not extended to foundations or beam joints	**7.5%**	1.9%	**Prominent**
14	**ADD_RC_04**	Additional	Pancake collapse of multi-storey educational frame structure	**7.5%**	1.9%	**Prominent**
15	**ADD_RC_06**	Additional	Dislodging of weakly connected overhead water tank appendages	**5.7%**	1.9%	**Prominent**
16	**RC_RVS_7.3**	RVS Primer	Building has large or heavy cantilever balconies or sunshades.	**3.8%**	1.9%	**Prominent**
17	**RC_RVS_3.5**	RVS Primer	Building has RC frame with Floating Columns.	**3.8%**	1.9%	**Prominent**
18	**RC_POST_5.1**	RVS Primer	Building has sustained collapse of cantilevers, balconies, chimneys and parapets.	**3.8%**	1.9%	**Prominent**
19	**RC_POST_3.5**	RVS Primer	Building has main load resisting columns with shear cracks.	**3.8%**	1.9%	**Prominent**
20	**RC_POST_3.4**	RVS Primer	Building has Floating Columns with cracked supporting beams.	**3.8%**	1.9%	**Prominent**
21	**RC_POST_2.3**	RVS Primer	Building is resting on liquefied soil.	**3.8%**	1.9%	**Prominent**
22	**ADD_RC_10**	Additional	Insufficient connection between RC elevator core wall and adjacent framing	**3.8%**	1.9%	**Prominent**
23	**ADD_RC_11**	Additional	Dislodgement of precast roof panels atop precast frame due to inadequate connections	**1.9%**	1.9%	**Prominent**
24	**ADD_RC_09**	Additional	Shear cracks in cantilever stub beam supporting floating columns	**1.9%**	1.9%	**Prominent**
25	**ADD_RC_07**	Additional	Temporary steel section or wooden post propping of damaged open ground storeys	**1.9%**	1.9%	**Prominent**
26	**ADD_RC_01**	Additional	Inadequate lap length in longitudinal column reinforcement	**1.9%**	1.9%	**Prominent**
27	**ADD_GEN_02**	Additional	Failure of welded tie connections between prestressed concrete trusses	**1.9%**	1.9%	**Prominent**

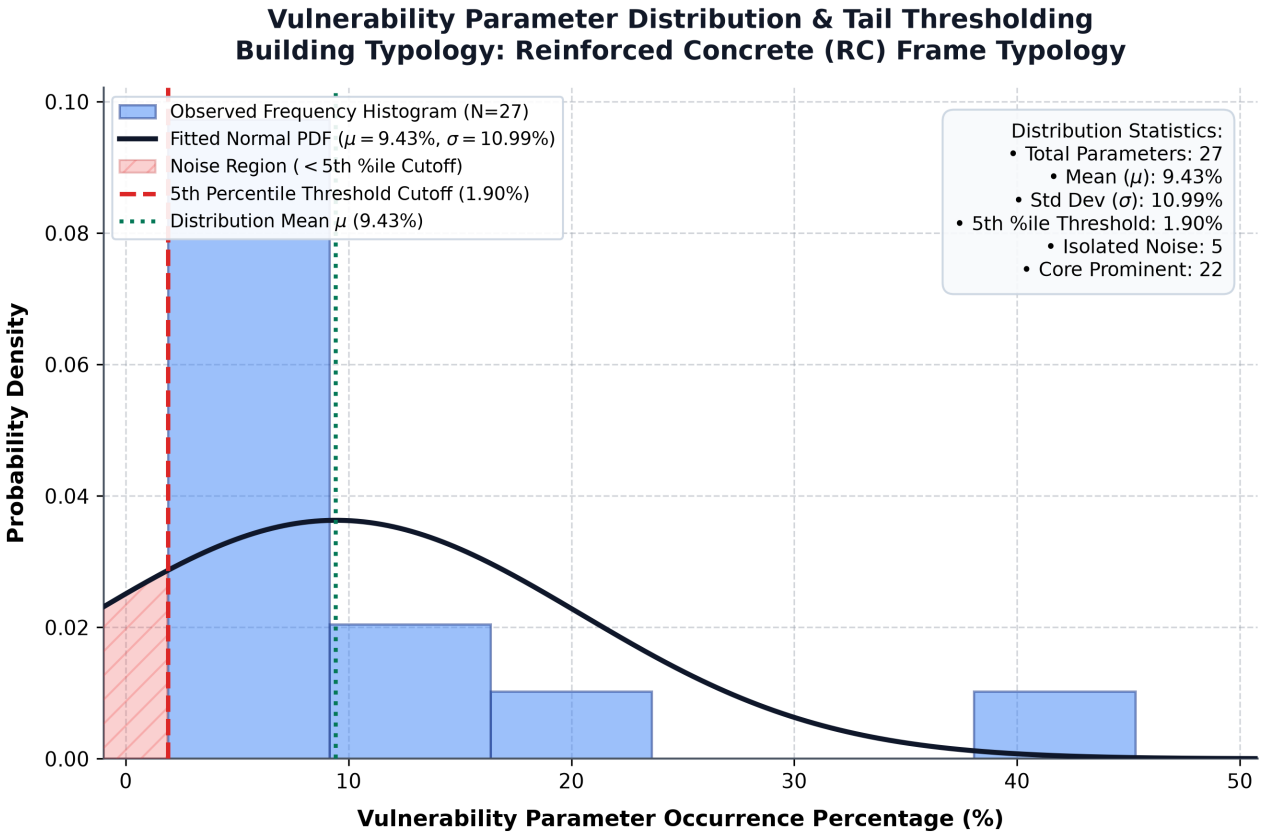


Figure: Normal distribution bell curve & 5% cutoff line for RC.

Typology: Random Rubble Stone Masonry (RRSM) Typology (RRSM)

Parameters Evaluated: 9 | **Mean (μ):** 6.67% | **Std Dev (σ):** 5.33% | **5% Threshold Cutoff:** 4.00% | **Core Prominent Retained:** 9

#	Parameter ID	Source	Vulnerability Description / Damage Mechanism	Occurrence	5% Cutoff	Status
1	**RRSM_POST_5.8**	RVS Primer	Building has floor-wall junction separated with walls out-of-plumb.	**20.0%**	4.0%	**Prominent**
2	**ADD_RRSM_01**	Additional	Out-of-plane collapse of stone masonry wall	**12.0%**	4.0%	**Prominent**
3	**RRSM_RVS_5.1**	RVS Primer	Building has walls made with mud mortar and deteriorated and cracked significantly owing to vagaries of outside weather (especially rainwater beating).	**4.0%**	4.0%	**Prominent**
4	**RRSM_RVS_4.6**	RVS Primer	Building has walls made with no or mud mortar exposed to vagaries.	**4.0%**	4.0%	**Prominent**
5	**RRSM_RVS_4.5**	RVS Primer	Building has masonry walls not integrated into each other at corners.	**4.0%**	4.0%	**Prominent**
6	**RRSM_RVS_4.3**	RVS Primer	Building has no continuous lintel band, roof band and gable band.	**4.0%**	4.0%	**Prominent**
7	**RRSM_POST_5.7**	RVS Primer	Building has suffered separation of wall junctions at corners.	**4.0%**	4.0%	**Prominent**
8	**RRSM_POST_5.4**	RVS Primer	Building has collapsed/damaged staircase/mumty or blockade of staircase.	**4.0%**	4.0%	**Prominent**
9	**ADD_RRSM_02**	Additional	Collapse of staircase and building corner in multi-storey stone masonry	**4.0%**	4.0%	**Prominent**

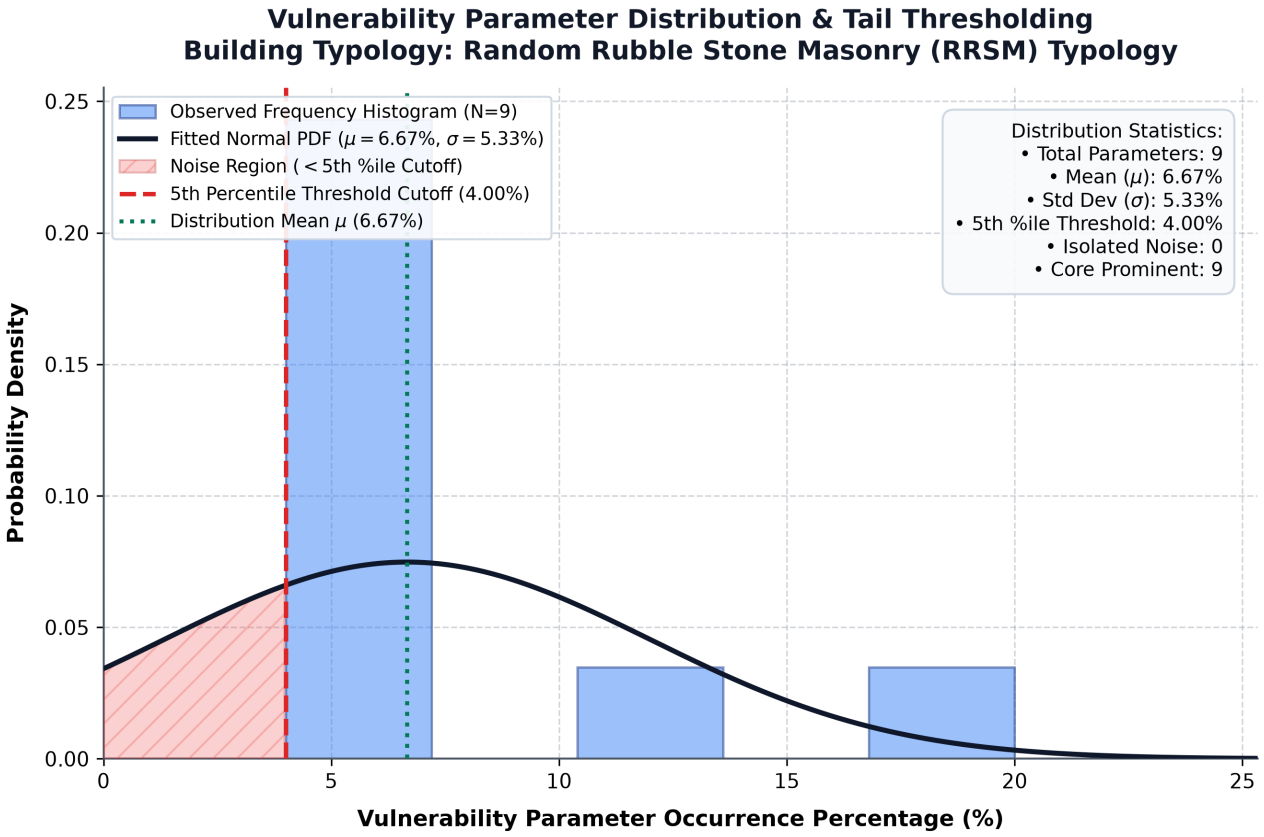


Figure: Normal distribution bell curve & 5% cutoff line for RRSM.